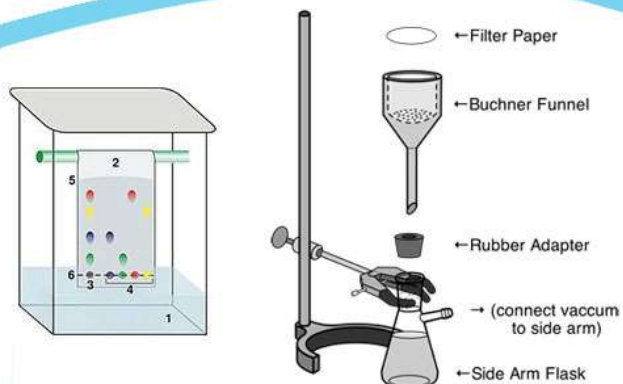


Subject: Environmental Sciences

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Paper No: 2 Analytical Chemistry

Module: 27 Polarography



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Description of Module	
Subject Name	Environmental Sciences
Paper Name	Analytical Chemistry
Module Name/Title	Polarography
Module Id	EVS/AC-II/27
Pre-requisites	
Objectives	1. What is polarography? 2. Why is it required? 3. Where and how it is used? 4. What is dropping mercury electrode? 5. How different polarography methods are grouped? 6. What is its importance?
Keywords	

Module 27: Polarography

Objectives: **To study the basics of polarography and know the following about self-generated questions.**

1. What is polarography?
2. Why is it required?
3. Where and how it is used?
4. What is dropping mercury electrode?
5. How different polarography methods are grouped?
6. What is its importance?

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MODULE 27: Polarography

Description:

It is a branch of voltammetry in which working electrode is dropping mercury electrode called microelectrode and a reference electrode mercury pool is used. The electrolyte consists of dilute solution of active species to be determined. Active species means which reacts on electrode. Along with the base electrolyte for furnishing good current is used. On passing variable potential (current-voltage) curve is obtained called polarogram and phenomenon is called polarography. Mercury pool in presence of chemical active species using a bulk of base electrolyte to study current voltage is called polarography.

Dropping mercury electrode: It is a working micro electrode and consist of pure Hg (purified by dilute HNO_3) to which a rubber tubing is attached bearing a fix glass capillary of 10-15 cm length having uniform bore with internal diameter 0.5 mm. By adjusting the height of reservoir the relative flow is adjusted. Usually a drop is formed between 1-5 seconds. It acts as a cathode and is attached along the negative terminal of battery through recorder. At anode it consists of Hg pool in contact will be a base electrolyte to which serve as reference electrode. If need is there external S.C.E may be used. It is attached to a positive terminal through potentiometer. Both cathode, anode are connected also to digital voltammeter. It is advisable that before passing current to expel oxygen dissolved in water by bubbling nitrogen or some other inert gas as interferes with C-V curve. On applying a variable potential to electrolyte current is measured at different interval of changing voltage. A plot is obtained between two to get a polarogram. A number of other micro electrodes have been used for the determination of current-voltage curve but the most satisfactory is a slow growing of dropping mercury electrode. The dropping mercury electrode has numerous advantages:

- (a) The surface of an electrode is conductive, smooth and continuously renewed to give better reproducibility of the current potential curve.
- (b) The steady value of diffusion current is reproducible even after the each change of an applied potential.
- (c) The mercury drop weights can be used for the calculation of surface area.

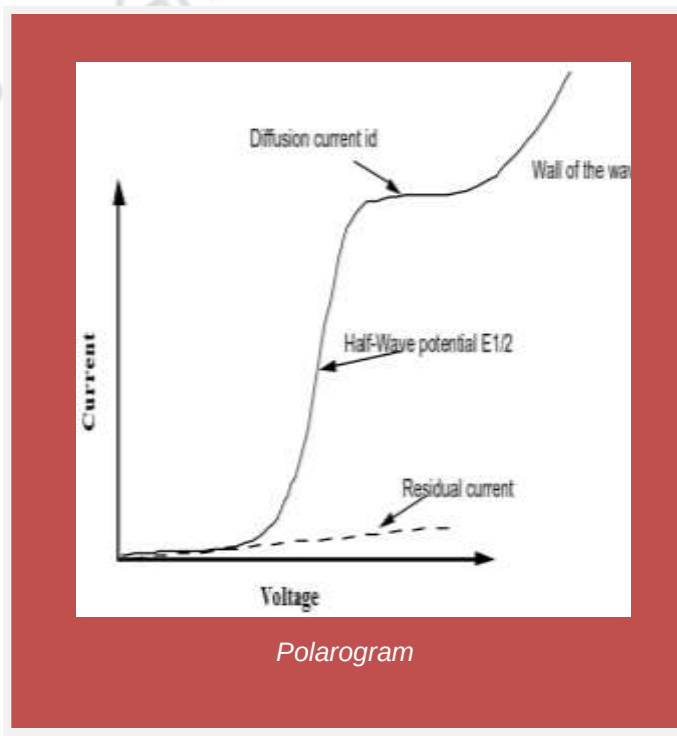
- (d) Many metal ions from amalgams with mercury.
- (e) No contamination at the surface of dropping mercury electrode occurs.
- (f) No poisoning of dropping mercury electrode occurs.
- (g) Since hydrogen has overvoltage with respect to mercury (SCE) a large number of metallic ions can be reduced.
- (h) The optimum range from dropping mercury electrode is 0.4 to 2.66 and 0.4 to -2.66.
- (i) Moreover after the reduction many metal get dissolves in pool of mercury to form amalgam which doesn't interfere in estimation.

Limitations

- (a) Due to the formation of mercury drop gradually surface area increases and consequently a little fluctuation in current may occur like an oscillation which may interferes estimation as an average current is considered.
- (b) DME can act as a good electrode between 0.4 to -2.66 V.

Basic Principle of polarography

All the factors based on theoretical concept of polarography influence the basic principle of polarography. A polarogram or a polarographic wave has represented. The slowly increasing current at the foot is known as residual current. In polarography it is necessary to use the inert base or non redox electrolyte for the correct estimation of active species. In case of only KCl (non redox electrolyte) after sweeping oxygen completely is taken a small amount of current is called as residual current is observed inspite of the fact there is no applied external voltage. This current is non-faradic in nature and arises due to the formation of electrically double layer. The



formation of double layer entirely depends upon the potential to be applied on dme either in the positive or in a negative range. With the passage of time when potential is gradually varies before decomposition is reached a little variation in the residual current with passage of time. More dme drop falls in solution of base electrolyte more double layers formation more residual current increases. So when there is increase in potential it is sufficient to cause electrochemical reaction for the base electrolyte. Again C-V curve for KCl etc can be seen beyond -2.66 V. Sometimes base current may increase due to the other impurity present in the base electrolyte. Furthermore, the migratory current is the current observed due to the migration of ions in the solution. Migration of ions occurs under the electrostatic electric field applied which may cause little variation in the current etc. The ions may diffuse quickly once the ions have reached the respective electrode after discharging conc. gradient (low conc.) at surface electrode setup. Migratory current is due to the migration of active ions under the potential difference applied to the cell. By the diffusion current arises due to the movement of ions under practically negligible emf of the cell. It is difficult to achieve negligible voltage in the cell. So it is advisable to use excess of supporting electrolyte (non redox) in nature which will nullify the charge on electrode under such condition active ions diffuse and reach the electrode. The current is exclusively the diffusion current. It is denoted by I_d . The diffusion current is determined by ilkovic equation:

$$I_d = 607nD^{1/2}cm^{2/3}t^{1/6}$$

Where, n = number of faraday produced for active species

C = concentration of electrolyte millimole/litre

D = Diffusion coefficient and is measured as cm^2/sec

m = the amount of mercury falling per second

t = time period of the drop in seconds

Ilkovic equation is some correction on RHS due to diffusion of ions of active species arising from the curvature of dropping mercury electrode. It is given by here the mass of falling drop

$$1 + AD^{1/2}tm^{-1/3}$$

where, A is attraction constant

m is mass of falling mercury drop

D is diffusion coefficient

Factors on which diffusion current depends:

- Temperature also effects the I_d as the ionic mobility is changed. For most of cases the temperature coefficient lies in between 1.5-2%. To control the temperature it is advisable to keep active electrolyte in a bath having a constant temperature about 25 °C.
- Pressure also influences the I_d either by changing the mass flow of mercury and affecting its speed. This can be controlled by adjusting the height of reservoir and adjusting the speed.
- Time alone factor doesn't affect the diffusion current because it has sixth root power coupled with mass flow, the diffusion current is affected. So, by choosing the size of capillary hole and adjusting the pressure one can get accurate result of I_d .
- Complex formation diffusion current is also greatly affected if active species from the complex in the solution. In case anionic complex or cationic complex is formed the movement of metal ion will be different i.e. hydrated complex will show different I_d than other complexes formed.
- Interfacial surface tension the diffusion current is effected by interfacial tension at mercury surface i.e. there is some gap between the surface of dme and solution phase containing active ions. As a result the current may decrease by decreasing the size of droplets as its interfacial tension can be reduced.
- Diffusion coefficient diffusion coefficient is related to both viscosity and concentration of the electrolyte. Viscosity depends on temperature and concentration of active electrolyte. If concentration of solution is low the viscosity will be low and diffusion current high. Alternatively, by increasing the temperature one can have high I_d . An empirical relation between I_d and η has been derived from the diffusion coefficient has been derived from stoke's Einstein formula:

$$D = RT/6\pi\eta Kr$$

Where R is gas constant

T = temperature

Π = Circle constant

η = Coefficient of viscosity

Since all other factors except η depend on the Brownian movement.

Viscosity of solution can be decreased not only by changing the concentration of active ion but diluting it with other solvent (acetonitrile, ethanol) etc. by the addition of alcohol initially 25% I_d is increased because alcohol makes complex with metal ion which is more lyophilic at mercury surface reduces the interfacial tension current will be increased.

Polarographic Maxima

Under the ideal conditions of concentration of active ion species as well as concentration of supporting electrolyte. The polarogram gives normal C-V curve is not obtained properly. Instead an abnormal increase in current is obtained which with passage of time decreases to give plateau. This abnormal increase in height of C-V curve is called polarographic maxima. These maxima may also arise besides the low concentration, pressure on dropping mercury electrode. These may vary in shape from sharp peaks to rounded heaps which gradually decrease to the normal diffusion current curve as the applied voltage is increased. But for the measurement of exact diffusion current the value of polarographic maxima must be eliminated or suppressed. The maxima are of two types:

Strike streaming type mechanism: Such type of maxima arises due to low concentration of base electrolyte and high concentration of active species. The unique feature of such mechanism maxima is that it occurs in a very short range of applied potential to dropping mercury electrode. Due to the low concentration of base electrolyte this leads to streaming movement of CV ions which are quickly diffused due to low viscosity and high rate of diffusion. The maxima appearing in CV curve causes the difficulties in measuring the diffusion current, concentration of active electrolyte. However such type of maxima can be controlled by decreasing the concentration of active ions and increasing the concentration of a base electrolyte.

Non streaming Maxima: Such type of maxima doesn't arising from streaming effect. But due to increase in pressure on reservoir attached to dropping mercury electrode the concentration of ions are

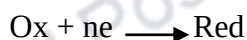
reduced by the quick fall of mercury drop which will give rise to broad maxima. $E_{1/2}$ and I_d are not measured properly. This can be controlled or eliminated by controlling the pressure or adding surface active material. To know the exact $E_{1/2}$, I_d etc there maxima must be eliminated by adding surface active material, capillary active ions, some surfactants or dyes etc.

Half wave potential

It is the potential of active ions when the diffusion current is half of the total current if diffusion current is I_d . At $E_{1/2}$ diffusion current is half. $E_{1/2}$ is the characteristic feature of the element and thus it gives the qualitative analysis, knowing the value of $E_{1/2}$ one can predict the element present in the solution.

In the usual current-voltage polarogram the total I_d current is given by the difference in height of plateau and base line current. By the convection at cathode reduction of oxidant (active ion M^{n+}) and is taken as the positive value while for the anode it is negative.

Before the start of the experiment the concentration of active ions in the bulk as well as near dme surface is same. Sometimes when the voltage is increased decomposition of active ion or Ox this reduced component moves in the bulk of solution.



Since the redox phenomenon is taking place in polarography. So Nernst equation is helpful in determining the potential and current of the redox system.

$$E = E^0 + \frac{RT}{nF} \ln \frac{a_{\text{ox}}}{a_{\text{red}}}$$

R = gas constant

a = activity

F = Faradays constant

n = no. of electrons involved during the redox

When the oxidant is reduced the other remaining active ions in the bulk will again diffuse to be discharged. Under such conditions current (I) at any point can be described on the basis of kinetics. For the sake of convenience the species present at dme surface which gets reduced (O_{xs}) stands for

surface the concentration gradient (O_x is a initial concentration and O_{xs} is after reduction). So the change in concentration is

$$I = [O_x] - [O_{xs}]$$

$[O_{xs}]$ is the amount reduced and moves to the bulk. So at dme concentration of $[O_{xs}]$ is zero. Under such conditions ions will diffuse and current is called diffusion current.

$$I = k[(O_x) - (O_{xs})] \dots \dots \dots (i)$$

$$I_d = k[O_x] \dots \dots \dots (ii)$$

From (i) and (ii)

$$I = I_d - k[O_{xs}]$$

$$[O_{xs}] = I_d - I / K$$

Significance of $E_{1/2}$ and I_d

$E_{1/2}$ tells us the qualitative aspect knowing $E_{1/2}$ one can predict the elements. By knowing the $E_{1/2}$ we can calculate the half of diffusion current. Knowing $E_{1/2}$ we know $I_{d/2}$ by doubling it. It will give total current which is proportional to the concentration of active. So one can know the quality of an electrolyte it is a qualitative aspect. So the total concentration of an active ion is directly proportional to the total current.

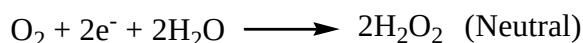
Quantitative technique and evaluation of result

Polarographic technique by for the best technique and result obtained by this technique are far superior than chemical technique and existing electroanalytical technique. It can be used to estimate the concentration ranging from 10^{-5} - 10^{-2} M. however the best results are obtained if the concentration lies in the range 10^{-4} – 10^{-3} M.

Determination of dissolved oxygen in water

Dissolved oxygen can be determined by chemical technique like $K_2Cr_2O_7$. The results may be tolerable but not upto the mark as demanded by the standard result. Since the range of concentration of dissolved oxygen falls in the prescribed range or limit of polarography. The polarographic technique

was used to estimate the amount of dissolved oxygen. If the polarogram of distilled water using the base electrolyte of suitable concentration usually 1 M and dye methyl orange 0.01 mM for the maxima suppressor. It is seen there occur reduction of oxygen in wide range of voltage. So, the current curve starts from -0.1 V and extend onwards. The first reduction potential at a low voltage is due to reduction of oxygen to H_2O_2 .



Here KCl is there which get adsorbed result in decreasing current. So for the purpose of compensating it acid is added. The second CV curve is present at high voltage in alkaline medium. This corresponds to reduction of H_2O_2 . Measuring the height of CV curve will give diffusion current and hence concentration directly proportional to diffusion current of dissolved oxygen can be evaluated fairly close to the standard value.

During the determination sometime mixture is given. In the mixture of electrolyte containing several ions if $E_{1/2}$ differ by tolerable limit then best estimation can be done. It is found that for M^+ difference in $E_{1/2}$ is 0.4. The best estimation in polarography technique can be obtained by using the masking reagent or making complex of interfering ions. As the C-V curve shifts to much negative reduction potential because the complex formation need high negative reduction potential to decompose. Thus the interfering CV curve is away and ion which to be estimated can be clearly determined. Thus Cu^{2+} can be complexed with KCN etc. Similarly Ag^+ can be best estimated in the presence of Pb^{2+} using SO_4^{2-} to form the ppt of Pb^{2+} to PbSO_4 having the no interference. How to measure the concentration of unknown electrolyte using the wave height

In this case several known amount of active ion solution are prepared by adding appropriate base electrolyte and maxima suppressor and their polarograms are obtained. The height between two plateau can be conveniently determined. A similar run under an identical condition is carried out for an unknown sample. Knowing the relative height of curve the result can be evaluated.

Internal addition of pilot ion

In the procedure the polarogram of an unknown sample, addition supporting electrolyte and maxima suppressor is taken. In the solution the pilot ion is added in a known amount and again

polarogram is taken. Measuring the height of a curve and comparing with first polarogram or from their ratio the value of an unknown can be found.

Standard addition Found

In this case the polarogram of an unknown sample is run. To this solution the standard solution of known strength of same ion is added. Then under identical condition from the relative height of polarogram one can find the amount of concentration of an unknown solution. This method is much better than the addition of pilot ion because as it may cause precipitation or maxima etc.

Let in the first case the current measured is I and the concentration of an unknown solution C_v according to illkovic equation

$$I_{1\infty} C_v$$

$$I = K C_v$$

$$K = I/C_v$$

Similarly for the second polarogram the known concentration C_s of volume v is taken and added to first solution and polarogram is run gives current I_2 .

$$I_2 = k [V C_v + V C_s]/(V+v)$$

$$K = I_2(V+v)/[V C_v + V C_s] \dots \dots \dots (i)$$

$$K = I_1/C_v \dots \dots \dots (ii)$$

Comparing (i) and (ii)

$$I_1/C_v = I_2(V+v)/(V C_v + V C_s)$$

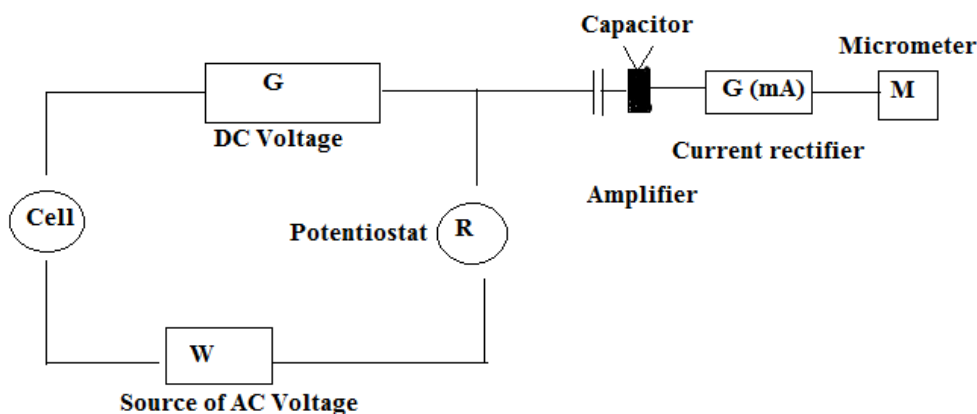
From all the values the C_v can be determined.

AC Polarography

A normal current voltage curve is obtained when the concentration of active electrolyte is minimum upto 10^{-4} M because at lower concentration curve is not defined well. Due to the changing size of mercury drop at DME current fluctuate which causes wrong estimation. Condenser current causes an error in the determination of I_d as it is non faradic in nature. Very dilute solution is not correctly estimated as resistance by solvent crops up. For the determination of mixture of ions one needs the $E_{1/2}$ potential difference 200 mV at least required otherwise there will be no clear separation of wave height.

Principle:

In order to have a clear separation of wave height for the different ions in solution and to know corrected I_d and $E_{1/2}$ etc. a modified method or modification in DC polarography is achieved by incorporating in alternative (AC) in the (DC) polarography. As AC polarography a constant sinusoidal current is fed to the cell containing active electrolyte with ac potential and dc potential superimposed on it. The purpose of dc is to give correct reduction/oxidation current in which there is no attraction/repulsion. Circuit diagram for AC polarogram is shown below:



Pulse Polarography

In pulse polarography, one can measure the changes in diffusion current produced by rectangular pulses of short duration. The pulse is synchronized with the maximum growth of mercury drop. If a dropping mercury electrode is employed then the current is measured from 40 to 60 milliseconds after the application of pulse to allow the time for the charging current to decay to a very low value. The capacitive current actually decays exponentially at a rate governed by magnitude of capacitance and series resistance of the system. During this time interval, the faradic current also decays somewhat but doesn't reach the diffusion control level because the concentration gradient at the instant of current measurement is considerably large. Each succeeding drop is polarized with somewhat large pulse.

The method gives a current-voltage curve similar to that of DC polarography and sensitivity of method is 6-8 V times than that of classical DC polarography. The measured signal is the faradic

current that flow at the plus potential and minus any faradic current flowing to the fixed DC potential. Types of pulse polarography are:

Normal pulse polarography:

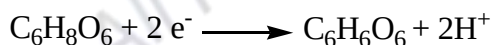
In this case pulses of gradually amplitude superimposed on a constant preset DC potential. So in the normal pulse polarography mode, the current potential relationship is relatively simple. If the initial potential is detected well before the rise or onset of the wave, the faradic current can be assumed to be equal to zero. The current potential curve for a reversible system is given by approximate equation:

$$I = nFCA\sqrt{D/\pi t_m} \times I/I+P$$

Where t_m is time interval between the pulse amplification and measured current. As the pulse potential becomes more negative than $E_{1/2}$ then P approached zero. So the limiting current given by:

$$I_l = nFCA D/\pi t_m$$

For example determination of ascorbic acid in conserved citrus juice. So ascorbic acid gives a well defined oscillation wave. So the determination can be carried out directly in freshly prepared citrus juice as well as conserved citrus juice.



Filtered the lemon or grape fruit through a porous funnel of size 1 mm. now prepare 25 mL for the graduate flask and add 0.5 mL acetate buffer, 2 mL of juice along with the standard addition of standard ascorbic acid and dilute it to the mark with distilled water.

Differential Pulse Polarography

In this case the superimposition of pulse of constant amplitude on a steadily constant D.C. potential. So the sensitivity is much better in which the current is sampled twice during a drop lifetime just before the drop. So the polarogram represents a current differences ΔI as a function of base plot then the curve is peak shape and height of peak is proportion to concentration of depolarizer. So the normal D.C. voltage ramp is applied to the system near the end of drop life and small amplitude pulse approximately 50 millivolts is superimposed on this range and the measured signal is the drift in the current measured before and after the amplification of current. So a change

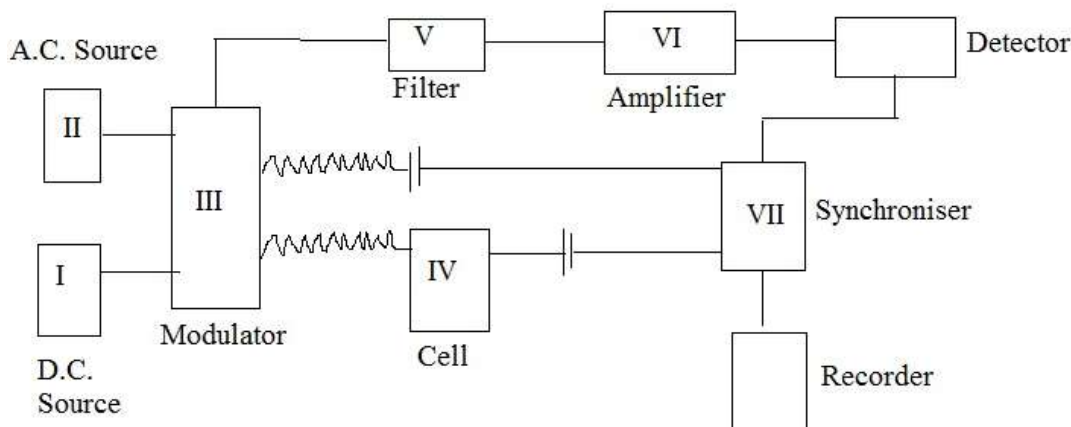
in current produces by the perturbation gives a peak shaped curve with a peak maxima occurring near $E_{1/2}$ wave potential if the perturbation is sufficiently small. For example in Pb^{2+} in 1 M KCl at the supporting electrolyte with both the ordinary and alternate drop, the decrease in charging current is accompanied by small diminished faradic current value with a pulse width of 0.4 msec drop time 1 sec $\Delta E = -50$ mV and the lead concentration as 40 to 160 nanogram per mL.

Radiofrequency polarography

It is a newly invented technique and has been claimed to be one of the most eloquent method of analysis. Although the known applications are still rare. Its theoretical bases are quite complex but conceptually it is simple. A sinusoidal radiofrequency 100 KHz -6.4 MHz signal w_1 and square wave modulated at 22.5 Hz w_2 superimposed on D.C. potential ramp. Then the response at 225 Hz is measured as in square wave polarography. Although barker presented this technique as a method of measuring D.C. rectification component in the presence of normal D.C. polarographic current and can be considered as a form of intermodulator polarography. The method therefore is a yet another second order technique resulting from non linearity of electrochemical cell. The applied wave form obviously includes a fourier component of different frequency. So the current observed at a particular frequency could be interpreted at the faradic intermodulation of the frequency with two side bands frequencies. The theoretical treatment using this approach has the same expression as those presented by Barker who observed that the current is same as that produce under the ordinary A.C. polarographic conditions when using a signal of peak to peak amplitude.

Square wave polarography

There is always some amount of double charge in case of A.C. or D.C. polarography which causes interference in diffusion current. The AC polarography is still causing much problem in double charge layer and on it changing charge on the sine wave which gives rise to inaccuracy in measuring the diffusion current. In A.C. polarography upto 10^{-5} M concentration results are fairly good but in advance square wave polarography if it is reversible system extra/high sensitivity in the instrument is needed. 10^{-6} M in an irreversible system accurate measurement can be achieved.



Principle:

Square wave polarograph technique is based on the application of square wave, low voltage of AC type superimposed on dc voltage. This method is quite successful in measuring mainly the faradic current due to the redox reaction and the least charging, double layer current etc. maximum amount of non faradic charging current has been eliminated. This technique involves higher sensitivity than the normal polarograph. The current is mainly measured at the end of drop where capacitor current is supposed to be zero. The instrument consists of it is a dc potential source applied to the system. It is a source square wave polarography ac current with potential having frequency of sin wave less than 200 Hz. Modulator for square wave polarography and dc current wave are modulated superimposed which are fed to cell and after that it is fed to synchronizer. Where c is the capacitor which tends to reduce the charging current etc. from the modulator current is filtered, amplify and fed to detector which is attached to synchronizer. The resultant current after addition, subtraction, division or subtraction of results various paths are shown in circuit. From the synchronizer is fed to recorder to record the square wave polarogram.

Applications of Polarography

Polarographic analyses can be used directly for the determination of any substance solid, liquid, or gaseous, organic or inorganic, ionic or molecular that can be reduced or oxidized at dropping mercury electrode. One of the most important advantages of polarography is the determination of two or more substances by obtaining a single current-potential curve. In addition to analytical uses,

polarography is one of the most fruitful techniques of research in physical, inorganic and organic chemistry. This technique is spreading more widely in subsidiary fields like biochemistry, pharmaceutical chemistry, environmental chemistry and others. It is that branch of Voltammetry in which changes in current, resulting from the electrolysis of the solution under study are investigated using a renewable mercury droplet as the indicator electrode (cathode). The anode of the electrolytic cell called the reference electrode consists of either a mercury pool at the bottom of the cell or a calomel electrode. The electro chemical technique Polarography used in analytical chemistry, involves measurements of current-voltage curves, obtained when voltage is applied to electrodes immersed in the solution being investigated. One of the electrodes is an indicator electrode. It is a dropping mercury electrode, consisting of a mercury drop hanging at the orifice of a fine bore glass capillary. Here the various applications of polarography are illustrated in table 1.

Table 1 simple examples illustrating contribution of polarography in interdisciplinary fields

Pharmaceutical science	• Separation based techniques are widely used in the analysis of drugs. • Analysis of tablets or injection solutions
Environmental chemistry	• Analysis of water samples for the presence of contaminants • Analysis of air for the presence of pollutants
Food industry	• Analysis of copper, lead, iron, vitamic etc in various food stuffs.
organic compounds	• Qualitative and quantitative determination of organic compounds. • Structure validation and evaluation
Trace metal alloys	• Predominately used for the analysis of trace metal alloys, minerals, their ores etc.
Toxicology	• Analysis of components of earth and developemnt of metallurgical process
Clinical analysis	• Analysis of blood samples, hair samples etc • Detection of poisons in the samples

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